

1. WAP for representing the polynomial using an array. Take the input from the user and display it in appropriate format.

Source Code:

```
#include <stdio.h>

int main(){
    int n;
    printf("Enter the highest exponent: ");
    scanf("%d", &n);

    int p[n+1];
    for (int i=0; i<n+1; i++) p[i]=0;

    int e; char ch;
    do{
        printf("Enter the exponent: ");
        scanf("%d", &e);
        printf("Enter the coefficient: ");
        scanf("%d", &p[e]);
        printf("Do you have another term? (y/n): ");
        scanf(" %c", &ch);
    } while (ch=='y');

    printf("Entered Polynomial: ");
    for (int i=n; i>=0; i--){
        if (p[i]==0) continue;
        if (p[i]>0) printf("+%dx^%d", p[i], i);
        if (p[i]<0) printf("%dx^%d", p[i], i);
    }

    printf("\n");
    return 0;
}
```

Output:

```
Enter the highest exponent: 5
Enter the exponent: 5
Enter the coefficient: 15
Do you have another term? (y/n): y
Enter the exponent: 3
Enter the coefficient: -20
Do you have another term? (y/n): y
Enter the exponent: 1
Enter the coefficient: 30
Do you have another term? (y/n): y
Enter the exponent: 0
Enter the coefficient: 90
Do you have another term? (y/n): n
Entered Polynomial: +15x^5-20x^3+30x^1+90x^0
```

2. WAP for addition of two polynomials using arrays.

Source Code:

```
#include <stdio.h>

int main(){
    int n;
    printf("Enter the highest exponent of P: ");
    scanf("%d", &n);

    int p[n+1];
    for (int i=0; i<n+1; i++) p[i]=0;

    int e; char ch;
    do{
        printf("Enter the exponent: ");
        scanf("%d", &e);
        printf("Enter the coefficient: ");
        scanf("%d", &p[e]);
        printf("Do you have another term? (y/n): ");
        scanf(" %c", &ch);
    } while (ch=='y');

    printf("P : ");
    for (int i=n; i>=0; i--){
        if (p[i]==0) continue;
        if (p[i]>0) printf("+%dx^%d", p[i], i);
        if (p[i]<0) printf("%dx^%d", p[i], i);
    }
    printf("\n");

    int m;
    printf("Enter the highest exponent of Q: ");
    scanf("%d", &m);

    int q[m+1];
    for (int i=0; i<m+1; i++) q[i]=0;

    do{
        printf("Enter the exponent: ");
        scanf("%d", &e);
        printf("Enter the coefficient: ");
        scanf("%d", &q[e]);
        printf("Do you have another term? (y/n): ");
        scanf(" %c", &ch);
    } while (ch=='y');

    printf("Q : ");
    for (int i=m; i>=0; i--){
        if (q[i]==0) continue;
        if (q[i]>0) printf("+%dx^%d", q[i], i);
        if (q[i]<0) printf("%dx^%d", q[i], i);
    }
}
```

```

printf("\n");

int size;
if (m < n) size = n;
else size = m;

int R[size+1];
for (int i=0; i<size+1; i++){
    if (size == n) R[i] = p[i];
    else R[i] = q[i];
}

if (size == n){
    for (int i=0; i<m+1; i++) R[i] += q[i];
} else {
    for (int i=0; i<m+1; i++) R[i] += p[i];
}

printf("Resultant Polynomail:");
for (int i=size; i>=0; i--){
    if (R[i]==0) continue;
    if (R[i]>0) printf("+%dx^%d", R[i], i);
    if (R[i]<0) printf("%dx^%d", R[i], i);
}
printf("\n");

return 0;
}

```

Output:

```
Enter the highest exponent of P: 5
Enter the exponent: 5
Enter the coefficient: 15
Do you have another term? (y/n): y
Enter the exponent: 3
Enter the coefficient: -20
Do you have another term? (y/n): y
Enter the exponent: 1
Enter the coefficient: 30
Do you have another term? (y/n): y
Enter the exponent: 0
Enter the coefficient: 90
Do you have another term? (y/n): n
P : +15x^5-20x^3+30x^1+90x^0
Enter the highest exponent of Q: 4
Enter the exponent: 4
Enter the coefficient: 4
Do you have another term? (y/n): y
Enter the exponent: 3
Enter the coefficient: 15
Do you have another term? (y/n): y
Enter the exponent: 1
Enter the coefficient: -10
Do you have another term? (y/n): y
Enter the exponent: 0
Enter the coefficient: 100
Do you have another term? (y/n): n
Q : +4x^4+15x^3-10x^1+100x^0
Resultant Polynomail:+15x^5+4x^4-5x^3+20x^1+190x^0
```

3. WAP for multiplication of two polynomials using arrays.

Source Code:

```
#include <stdio.h>

int main(){
    int n;
    printf("Enter the highest exponent of P: ");
    scanf("%d", &n);

    int p[n+1];
    for (int i=0; i<n+1; i++) p[i]=0;

    int e; char ch;
    do{
        printf("Enter the exponent: ");
        scanf("%d", &e);
        printf("Enter the coefficient: ");
        scanf("%d", &p[e]);
        printf("Do you have another term? (y/n): ");
        scanf(" %c", &ch);
    } while (ch=='y');

    printf("P : ");
    for (int i=n; i>=0; i--){
        if (p[i]==0) continue;
        if (p[i]>0) printf("+%dx^%d", p[i], i);
        if (p[i]<0) printf("%dx^%d", p[i], i);
    }
    printf("\n");

    int m;
    printf("Enter the highest exponent of Q: ");
    scanf("%d", &m);

    int q[m+1];
    for (int i=0; i<m+1; i++) q[i]=0;

    do{
        printf("Enter the exponent: ");
        scanf("%d", &e);
        printf("Enter the coefficient: ");
        scanf("%d", &q[e]);
        printf("Do you have another term? (y/n): ");
        scanf(" %c", &ch);
    } while (ch=='y');

    printf("Q : ");
    for (int i=m; i>=0; i--){
        if (q[i]==0) continue;
        if (q[i]>0) printf("+%dx^%d", q[i], i);
        if (q[i]<0) printf("%dx^%d", q[i], i);
    }
}
```

```

printf("\n");

int size = m*n;

int R[size+1];
for (int i=0; i<size+1; i++) R[i] = 0;

for (int i=0; i<n+1; i++){
    for (int j=0; j<m+1; j++){
        R[i+j] = R[i+j] + (p[i]*q[j]);
    }
}

printf("Resultant Polynomail:");
for (int i=size; i>=0; i--){
    if (R[i]==0) continue;
    if (R[i]>0) printf("+%dx^%d", R[i], i);
    if (R[i]<0) printf("%dx^%d", R[i], i);
}
printf("\n");

return 0;
}

```

Output:

```

Enter the highest exponent of P: 3
Enter the exponent: 3
Enter the coefficient: 10
Do you have another term? (y/n): y
Enter the exponent: 2
Enter the coefficient: 5
Do you have another term? (y/n): y
Enter the exponent: 0
Enter the coefficient: 10
Do you have another term? (y/n): n
P : +10x^3+5x^2+10x^0
Enter the highest exponent of Q: 2
Enter the exponent: 2
Enter the coefficient: 15
Do you have another term? (y/n): y
Enter the exponent: 1
Enter the coefficient: 20
Do you have another term? (y/n): y
Enter the exponent: 0
Enter the coefficient: 30
Do you have another term? (y/n): n
Q : +15x^2+20x^1+30x^0
Resultant Polynomail:+150x^5+275x^4+400x^3+300x^2+200x^1+300x^0

```

4. WAP to create the triplet representation of a sparse matrix.

Source Code:

```
#include <stdio.h>

int main(){
    int r,c;
    printf("Enter the no. of rows and columns: ");
    scanf("%d %d", &r, &c);

    int A[r][c];
    printf("Enter the Elements:\n");
    for (int i=0; i<r; i++){
        for (int j=0; j<c; j++){
            printf("Element %d%d: ", i, j);
            scanf("%d", &A[i][j]);
        }
    }

    for (int i = 0; i < r; i++)
    for (int j = 0; j < c; j++)
    printf("%d ", A[i][j]);
    printf("\n");

    int count = 0;
    for (int i=0; i<r; i++)
    for (int j=0; j<c; j++)
    if (A[i][j] != 0)
    count++;

    if (((r*c)-count) >= ((r*c)/2)){

        int S[count+1][3];
        S[0][0] = r;
        S[0][1] = c;
        S[0][2] = count;

        int k=1;
        for (int i=0; i<r; i++)
        for (int j=0; j<c; j++)
        if (A[i][j] != 0){
            S[k][0] = i;
            S[k][1] = j;
            S[k][2] = A[i][j];
            k++;
        }

        for (int i = 0; i < count+1; i++){
            for (int j = 0; j < 3; j++)
                printf("%d ", S[i][j]);
            printf("\n");
        }
    }
}
```

```
    return 0;  
}
```

Output:

```
Enter the no. of rows and columns: 4 4  
Enter the Elements:  
Element 00: 0  
Element 01: 0  
Element 02: 5  
Element 03: 0  
Element 10: 0  
Element 11: 8  
Element 12: 0  
Element 13: 0  
Element 20: 0  
Element 21: 0  
Element 22: 0  
Element 23: 3  
Element 30: 2  
Element 31: 0  
Element 32: 0  
Element 33: 0  
0 0 5 0 0 8 0 0 0 0 0 3 2 0 0 0  
4 4 4  
0 2 5  
1 1 8  
2 3 3  
3 0 2
```

5. WAP to find the addition of two sparse matrices using triplet format.

Source Code:

```
#include <stdio.h>

int main(){
    int r, c;
    printf("Enter the no. of rows and columns: ");
    scanf("%d %d", &r, &c);

    int A[r][c];
    printf("Enter the Elements for Matrix A:\n");
    for (int i=0; i<r; i++){
        for (int j=0; j<c; j++){
            printf("Element %d%d: ", i, j);
            scanf("%d", &A[i][j]);
        }
    }

    int B[r][c];
    printf("Enter the Elements for Matrix B:\n");
    for (int i=0; i<r; i++){
        for (int j=0; j<c; j++){
            printf("Element %d%d: ", i, j);
            scanf("%d", &B[i][j]);
        }
    }

    int countA = 0;
    for (int i=0; i<r; i++)
        for (int j=0; j<c; j++)
            if (A[i][j] != 0)
                countA++;

    int countB = 0;
    for (int i=0; i<r; i++)
        for (int j=0; j<c; j++)
            if (B[i][j] != 0)
                countB++;

    if (((r*c)-countA) >= ((r*c)/2) && ((r*c)-countB) >= ((r*c)/2)){
        int SA[countA+1][3];
        SA[0][0] = r;
        SA[0][1] = c;
        SA[0][2] = countA;

        int k=1;
        for (int i=0; i<r; i++)
            for (int j=0; j<c; j++)
                if (A[i][j] != 0){
                    SA[k][0] = i;
                    SA[k][1] = j;
                    SA[k][2] = A[i][j];
```

```

        k++;
    }

    int SB[countB+1][3];
    SB[0][0] = r;
    SB[0][1] = c;
    SB[0][2] = countB;

    k=1;
    for (int i=0; i<r; i++)
        for (int j=0; j<c; j++)
            if (B[i][j] != 0){
                SB[k][0] = i;
                SB[k][1] = j;
                SB[k][2] = B[i][j];
                k++;
            }

    printf("Sparse Matrix A:\n");
    for (int i = 0; i < countA+1; i++){
        for (int j = 0; j < 3; j++)
            printf("%d ", SA[i][j]);
        printf("\n");
    }

    printf("Sparse Matrix B:\n");
    for (int i = 0; i < countB+1; i++){
        for (int j = 0; j < 3; j++)
            printf("%d ", SB[i][j]);
        printf("\n");
    }

    int SC[countA+countB+1][3];
    SC[0][0] = r;
    SC[0][1] = c;
    SC[0][2] = countA+countB;

    for (int i=1; i<=countA; i++){
        SC[i][0] = SA[i][0];
        SC[i][1] = SA[i][1];
        SC[i][2] = SA[i][2];
    }

    int t = countA + 1;
    for (int i=1; i<=countB; i++){
        SC[t][0] = SB[i][0];
        SC[t][1] = SB[i][1];
        SC[t][2] = SB[i][2];
        t++;
    }

    printf("Sparse Matrix Sum (3-Tuple):\n");
    for (int i = 0; i < t; i++){
        for (int j = 0; j < 3; j++)

```

```

        printf("%d ", SC[i][j]);
        printf("\n");
    }
}
return 0;
}

```

Output:

```

Enter the no. of rows and columns: 3 3
Enter the Elements for Matrix A:
Element 00: 0
Element 01: 0
Element 02: 5
Element 10: 0
Element 11: 8
Element 12: 0
Element 20: 2
Element 21: 0
Element 22: 0
Enter the Elements for Matrix B:
Element 00: 0
Element 01: 3
Element 02: 0
Element 10: 0
Element 11: 0
Element 12: 4
Element 20: 0
Element 21: 0
Element 22: 0
Sparse Matrix A:
3 3 3
0 2 5
1 1 8
2 0 2
Sparse Matrix B:
3 3 2
0 1 3
1 2 4
Sparse Matrix Sum (3-Tuple):
3 3 5
0 2 5
1 1 8
2 0 2
0 1 3
1 2 4

```

6. WAP to find the transpose of a sparse matrix using triplet format only.

Source Code:

```
#include <stdio.h>

int main(){
    int r,c;
    printf("Enter the no. of rows and columns: ");
    scanf("%d %d", &r, &c);

    int A[r][c];
    printf("Enter the Elements:\n");
    for (int i=0; i<r; i++){
        for (int j=0; j<c; j++){
            printf("Element %d%d: ", i, j);
            scanf("%d", &A[i][j]);
        }
    }

    printf("Original Matrix:\n");
    for (int i = 0; i < r; i++){
        for (int j = 0; j < c; j++)
            printf("%d ", A[i][j]);
        printf("\n");
    }

    int count = 0;
    for (int i=0; i<r; i++)
        for (int j=0; j<c; j++)
            if (A[i][j] != 0)
                count++;

    if (((r*c)-count) >= ((r*c)/2)){
        int S[count+1][3];
        S[0][0] = r;
        S[0][1] = c;
        S[0][2] = count;

        int k=1;
        for (int i=0; i<r; i++)
            for (int j=0; j<c; j++)
                if (A[i][j] != 0){
                    S[k][0] = i;
                    S[k][1] = j;
                    S[k][2] = A[i][j];
                    k++;
                }

        printf("Original 3-Tuple Matrix:\n");
        for (int i = 0; i < count+1; i++){
            for (int j = 0; j < 3; j++)
                printf("%d ", S[i][j]);
            printf("\n");
        }
    }
}
```

```

    }

    int T[count+1][3];
    T[0][0] = S[0][1];
    T[0][1] = S[0][0];
    T[0][2] = S[0][2];

    int t = 1;
    for (int col = 0; col < c; col++){
        for (int i = 1; i <= count; i++){
            if (S[i][1] == col){
                T[t][0] = S[i][1];
                T[t][1] = S[i][0];
                T[t][2] = S[i][2];
                t++;
            }
        }
    }

    printf("Transpose 3-Tuple Matrix:\n");
    for (int i = 0; i < count+1; i++){
        for (int j = 0; j < 3; j++){
            printf("%d ", T[i][j]);
            printf("\n");
        }
    }
    return 0;
}

```

Output:

Enter the no. of rows and columns: 3 3

Enter the Elements:

Element 00: 0

Element 01: 0

Element 02: 5

Element 10: 0

Element 11: 8

Element 12: 0

Element 20: 2

Element 21: 0

Element 22: 0

Original Matrix:

0 0 5

0 8 0

2 0 0

Original 3-Tuple Matrix:

3 3 3

0 2 5

1 1 8

2 0 2

Transpose 3-Tuple Matrix:

3 3 3

0 2 2

1 1 8

2 0 5